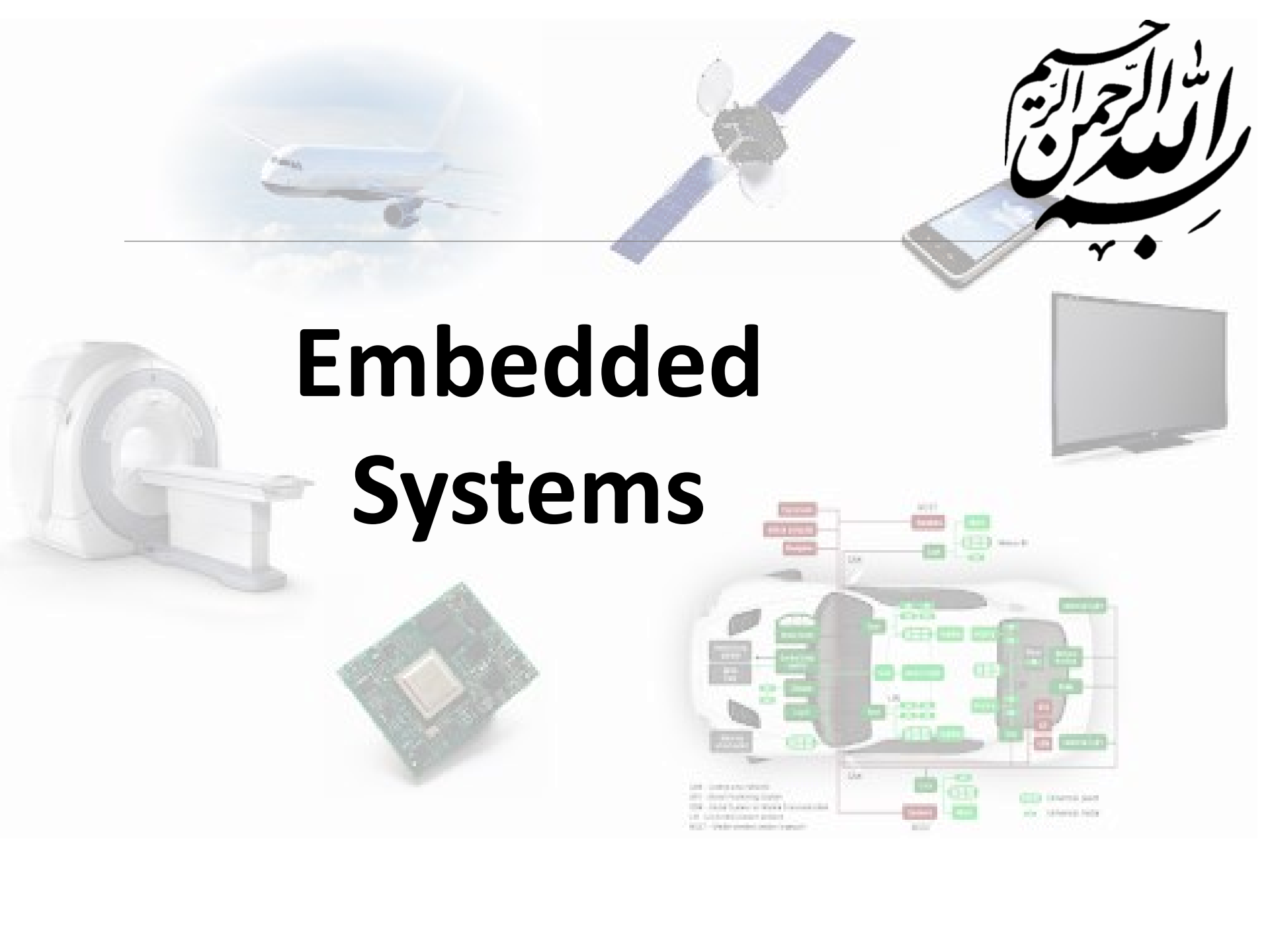


السلامة

Embedded Systems



Lecture2: Embedded System Components

Components-peripherals

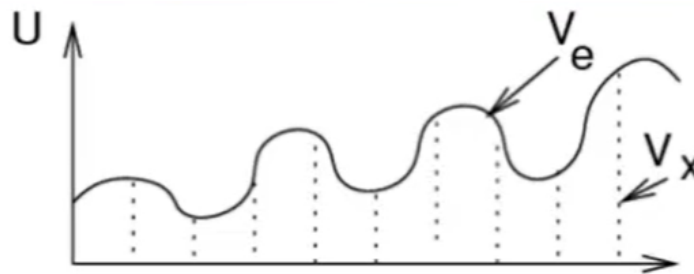
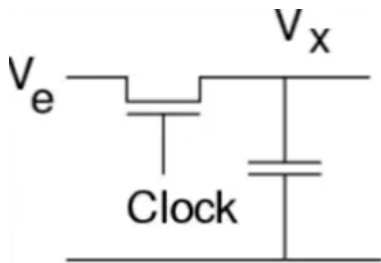
➤ **Perform specific computation task (ASIC)**

- Custom single-purpose processors
Designed by us for unique task
- Standard single-purpose processors
Off-the-shelf
Pre-designed for a common task

Components-peripherals

➤ Analog to digital conversion

- **Sampling:** how often is the signal converted?
- **Quantization:** how many bits used to represent a sample?



$$V_x = V_e \text{ when clock} = 1$$

Components-peripherals

➤ Timers (very important in RT and ES)

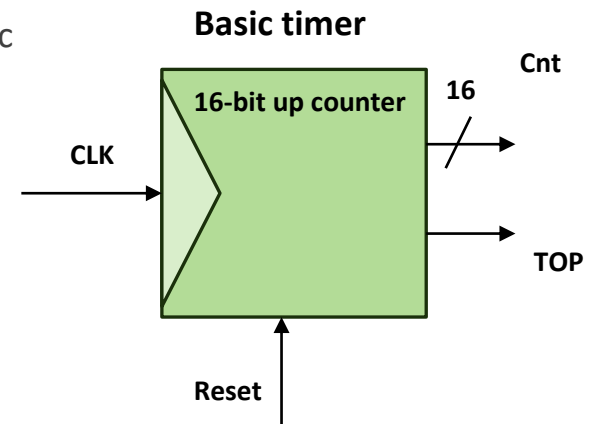
■ Measure time intervals

- To generate timed output events
- To measure input events
- Top: max count reached

■ Range and resolution (interval)

EX:

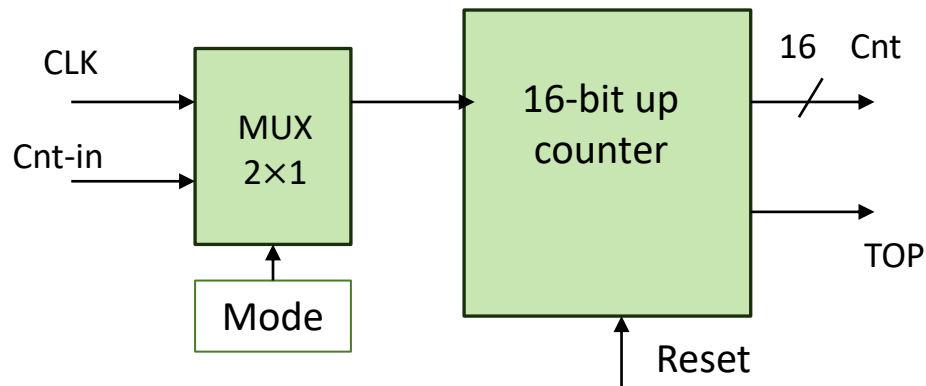
- Width of clock pulse: 1 s (1Hz) and 8 bits-timer → max:256 sec
- Width of clock pulse: 0.5 s (2 Hz) and 8 bits-timer → max: 128 sec



Components-peripherals

➤ Counters:

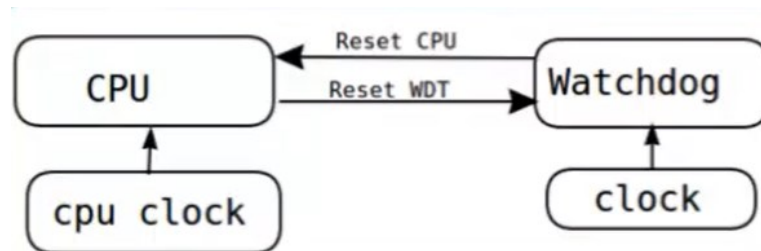
- Like a timer, but counts pulses on a general input signal rather than clock
 - E.g., counts cars passing over a sensor
 - Can often configure device as either a timer or counter



Components-peripherals

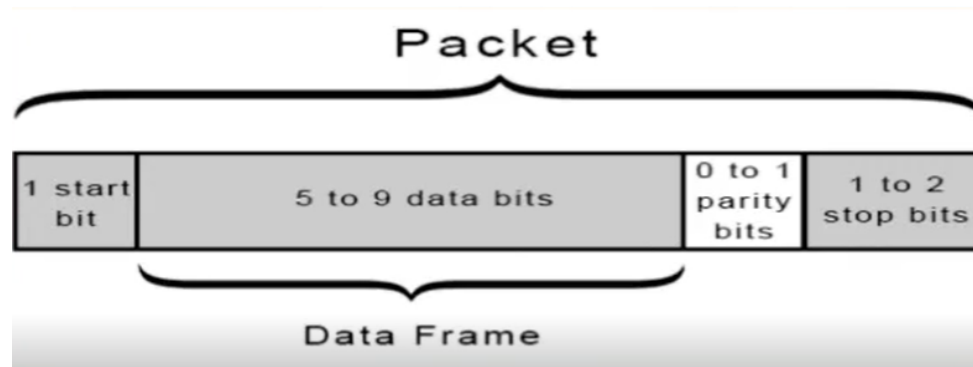
➤ Watchdog Timer

- Must reset timer every X time unit, else timer generates a signal
- Common use: detect failure, self-reset



Components-peripherals

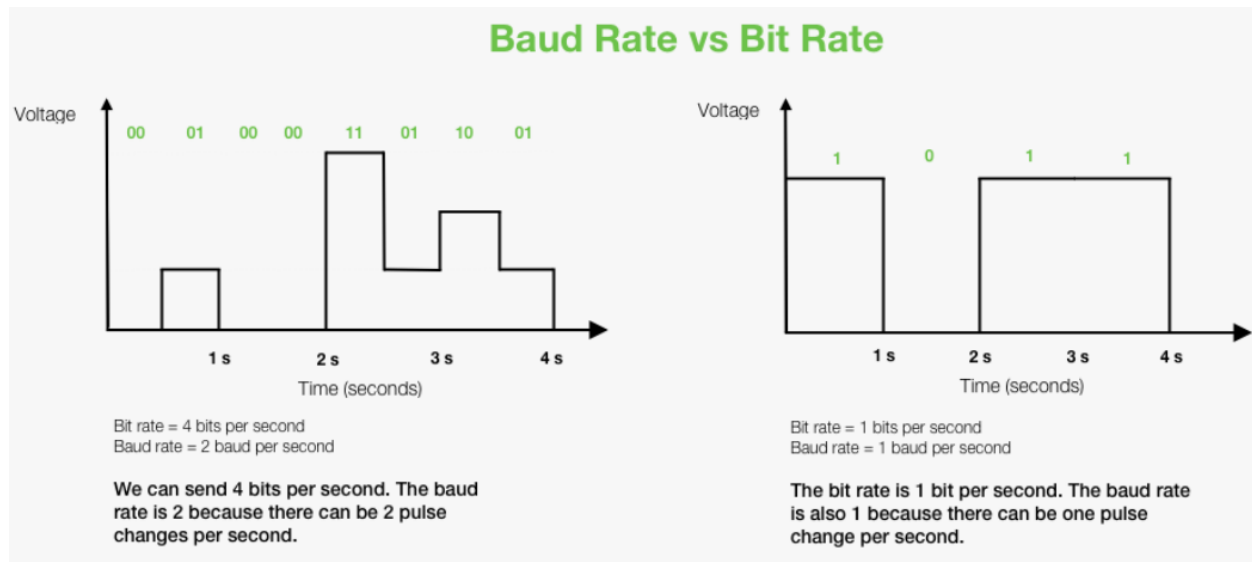
- **Universal Asynchronous Receiver Transmitter (UART):**
 - Takes parallel data and transmits serially (is close to processor)
 - Receives serial data and converts to parallel
 - Parity: extra bit for simple error checking
 - Start bit, stop bit



Components-peripherals

➤ UART-Baud rate

- Signal changes per second (sample/sec)
- Bit rate. Sometimes different



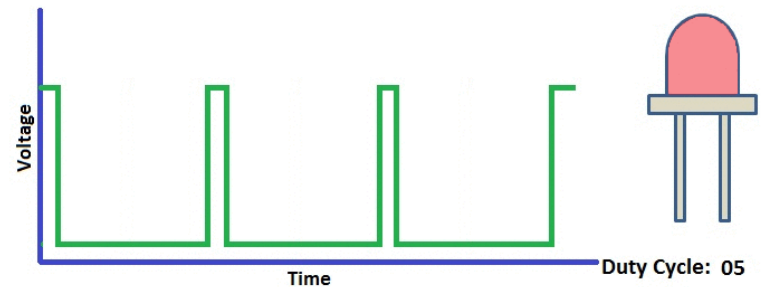
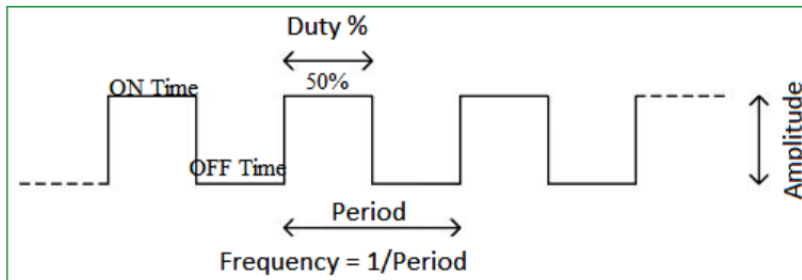
$$\text{Bit rate} = \text{Baud rate} \times \text{the number of bit per baud}$$

Components-peripherals

➤ Pulse width modulator (PWM)

- Generates pulses with specific high/low times
- Duty cycle: % time high
- Square wave: 50% duty cycle

Duty Cycle = Turn ON time / (Turn ON time + Turn OFF time)



Components-peripherals

➤ Pulse width modulator (PWM)

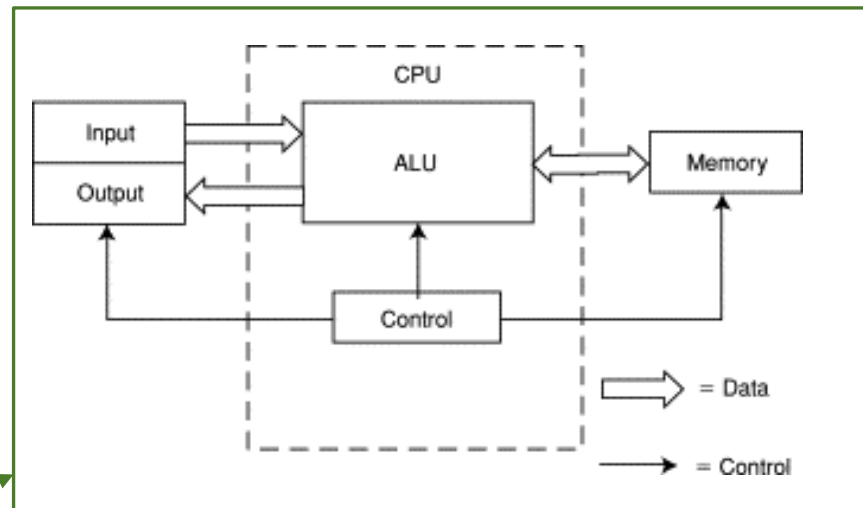
- Generates pulses with specific high/low times
- Duty cycle: % time high
Square wave: 50% duty cycle
- Common use: control average voltage to electric device
 - DC-DC converter or digital-analog converter
 - DC motor speed, dimmer lights

Microprocessors vs Microcontrollers

➤ Microprocessors vs Microcontrollers

Basic components of Digital computer

- CPU
- Memory
- I/O



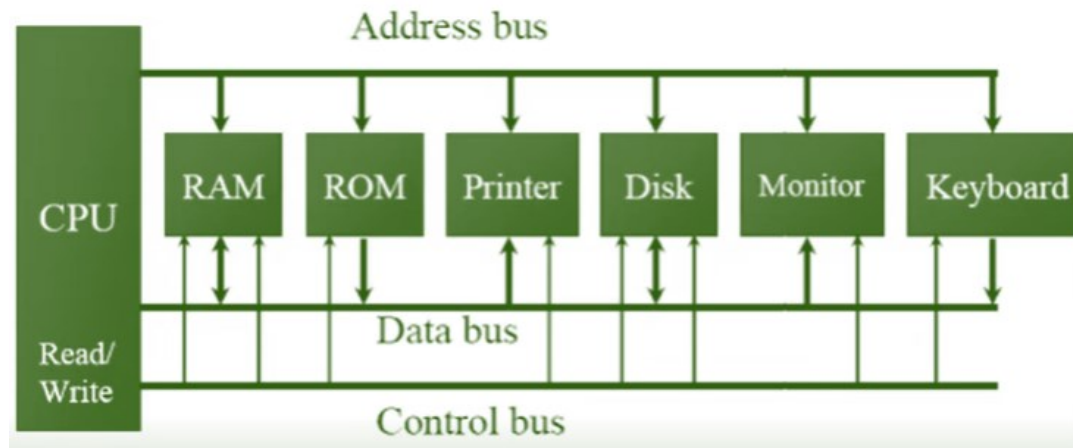
Could be a board, several boards

Microprocessors vs Microcontrollers

➤ Microprocessors vs Microcontrollers

- Commonly no RAM, ROM, I/O on CPU chip itself
- Many chips on motherboard

Microprocessor-based systems



Microprocessors vs Microcontrollers

➤ Microprocessors vs Microcontrollers

- Embedded systems: Features
 - Size of product
 - Power consumption
 - Price
 - ...

Microprocessors vs Microcontrollers

➤ Microprocessors vs Microcontrollers

- Microcontroller (MCU)

A small computer on a single integrated circuit

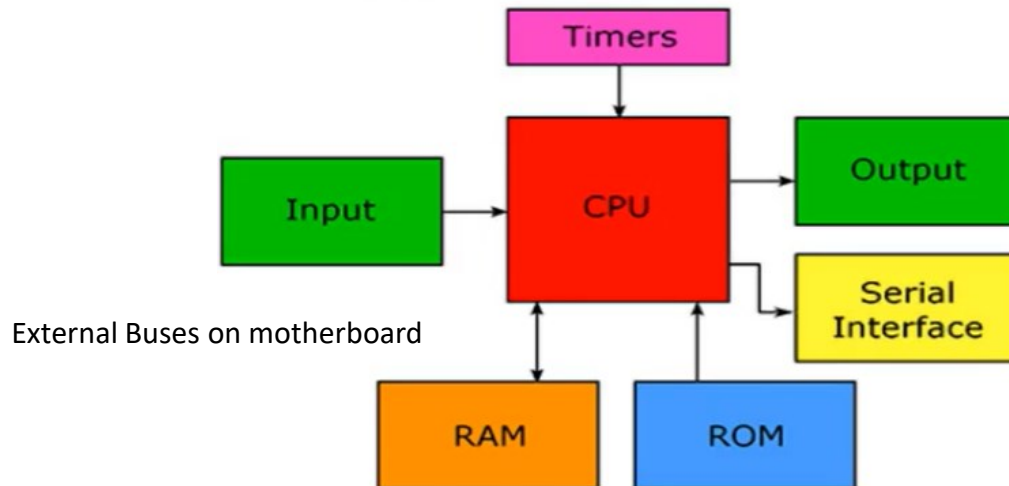
- On-chip RAM, ROM, I/O ports...



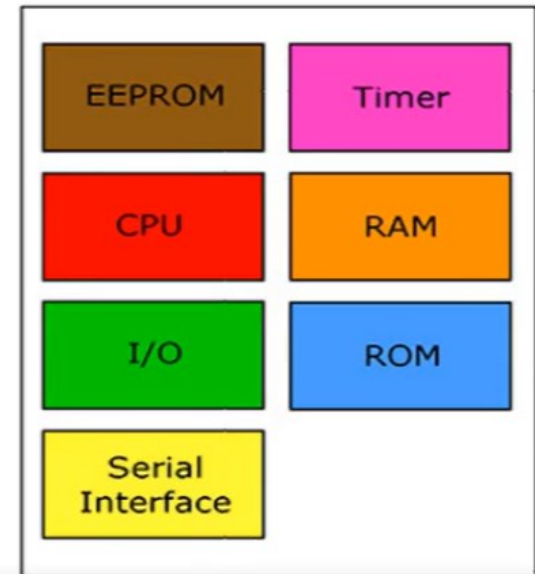
Microprocessors vs Microcontrollers

➤ Microprocessors vs Microcontrollers

Microprocessor: CPU and several supporting chips.



Microcontroller: CPU on a single chip.



Microprocessors vs Microcontrollers

➤ Microprocessors vs Microcontrollers

Microprocessor (MPU)

- CPU is stand-alone, RAM, ROM, I/O, timer are separate
- Designer can decide on the amount of ROM, RAM and I/O ports.
- Expensive
- General-purpose
- High power consumption
- Typically 32/64-bit
- Typically deep pipeline (5-20 stages)

Microcontroller (MCU)

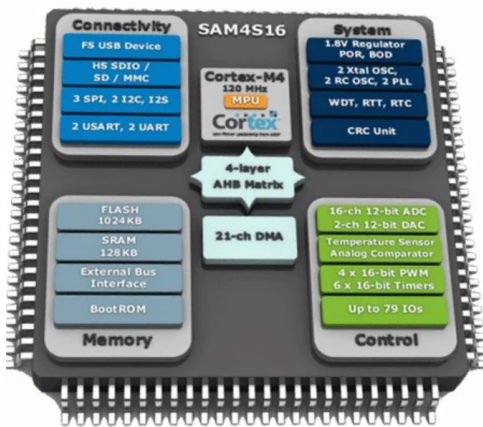
- CPU, RAM, ROM, I/O and timer are all on a single chip
- Fixed amount of on-chip ROM, RAM, I/O ports
- For applications in which cost, power and space are critical
- Single-purpose (control-oriented)
- Low power consumption
- Typically 8/16 bit
- Typically single-cycle/two-stage pipeline

Microprocessors vs Microcontrollers

➤ Microcontrollers vs system on chip (SOC)

Micro Control Unit (MCU)

- A small computer on single integrated circuit
- It is similar to, but less sophisticated than, a system on chip (SOC)



MCU

❖ Compared to the general-purpose microprocessors used in personal computers, an **MCU** emphasizes

1. **self-sufficiency** (not relying on external hardware)
2. **cost-effectiveness**
3. **Compact size**, allowing it to be placed within devices
4. **limited storage capacity**
5. **simple input/output interfaces**
6. **lower functionality**

- ✓ The development of MCUs has gone through phases including 8-bit 8051 microcontrollers, 16-bit AVR microcontrollers, and 32-bit STM microcontrollers.
7. MCUs integrate processors and various modules such as USB controllers, UART controllers, memory, Flash, etc.

Microprocessors vs Microcontrollers

➤ Microcontrollers vs system on chip (SOC)

Microcontrollers (MCU): A small computer on a single integrated circuit

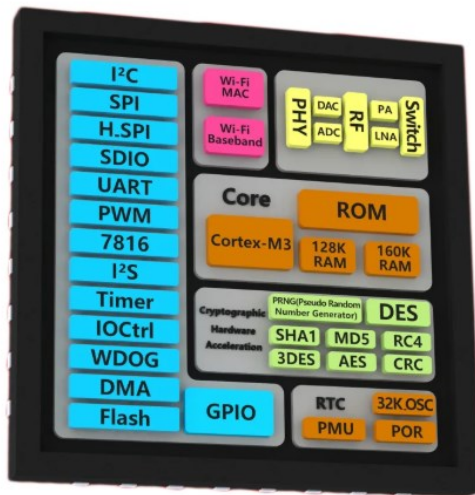
- A microcontroller contains:
- One or more CPUs (process core)
- Memory:
 - FLASH
 - RAM
 - ...
- Programmable input/output peripherals
- Microcontrollers are designed for embedded applications



Microprocessors vs Microcontrollers

➤ Microcontrollers vs system on chip (SOC)

System on chip (SOC)



Embedded Wi-Fi SoC Chip W600

❖ A System on a Chip (SOC) is an integrated circuit that integrates a computer or other electronic systems onto a single chip. To provide a comparison:

- An MCU is chip-level, while an SOC is system-level.
- It combines the advantages of MCUs and MPUs, having built-in RAM and ROM while also being powerful like an MPU.
- It can store and execute system-level code, meaning it can run operating systems (primarily Linux OS).

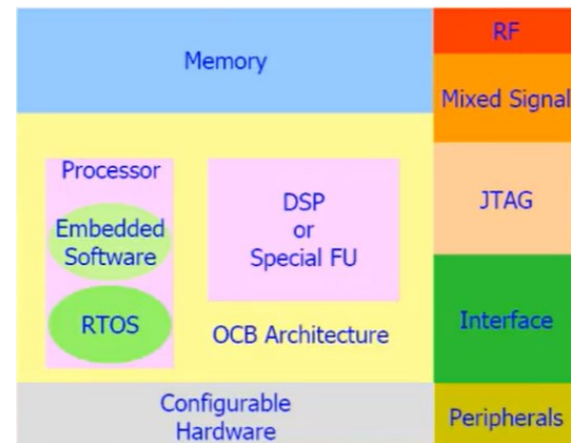
Microprocessors vs Microcontrollers

➤ Microcontrollers vs system on chip (SOC)

System on chip (SOC)

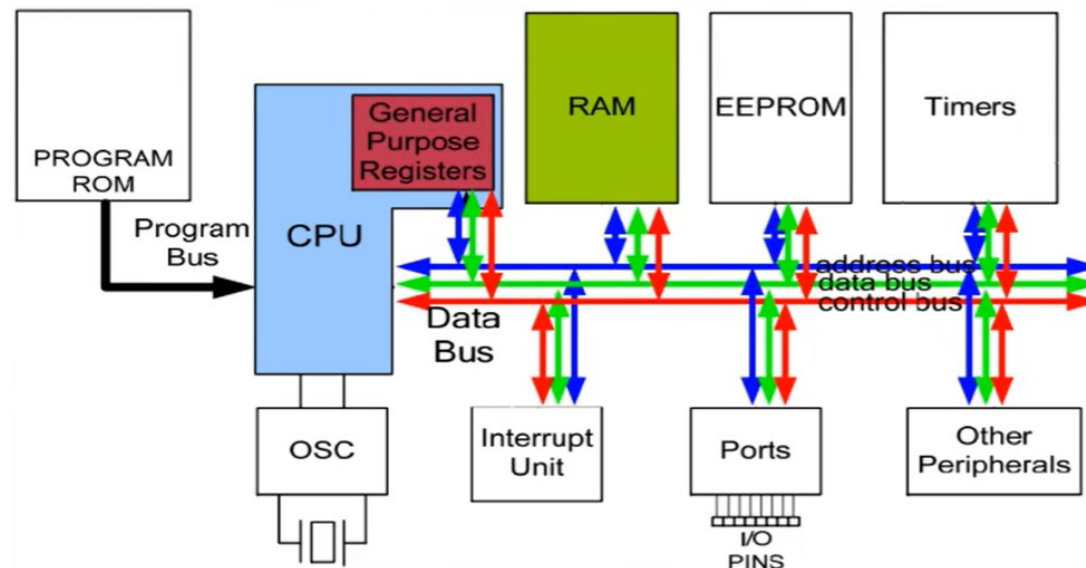
System on a Chip (SOC) may contain Custom hardware blocks

- DSP blocks
- Embedded CPU
- Embedded memory
- Real world interfaces (USB, PCI, Ethernet)
- Software (both OS and applications)
- Mixed-signal blocks
DAC, ADC



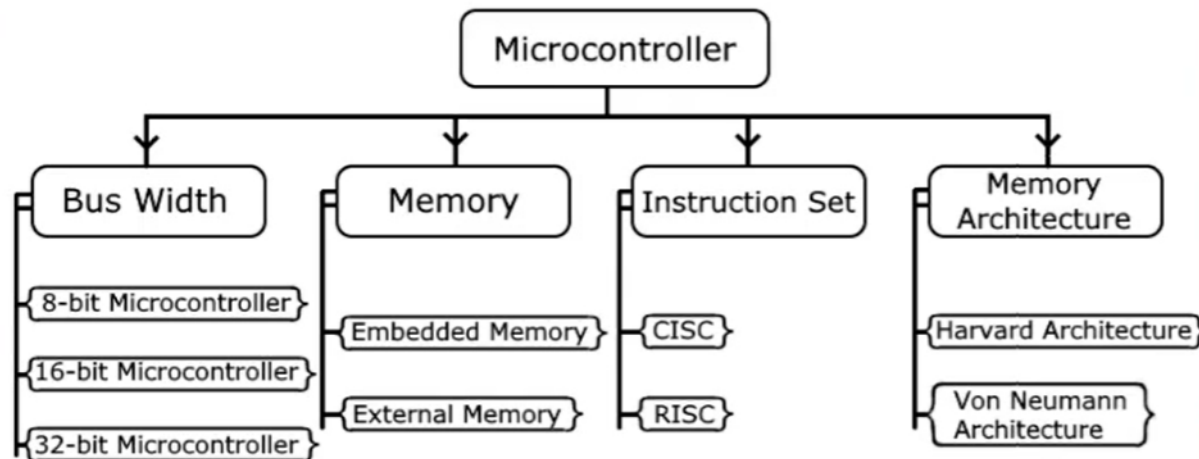
Microcontrollers

➤ Microcontrollers: General view



Microcontrollers

➤ Microcontrollers: Types

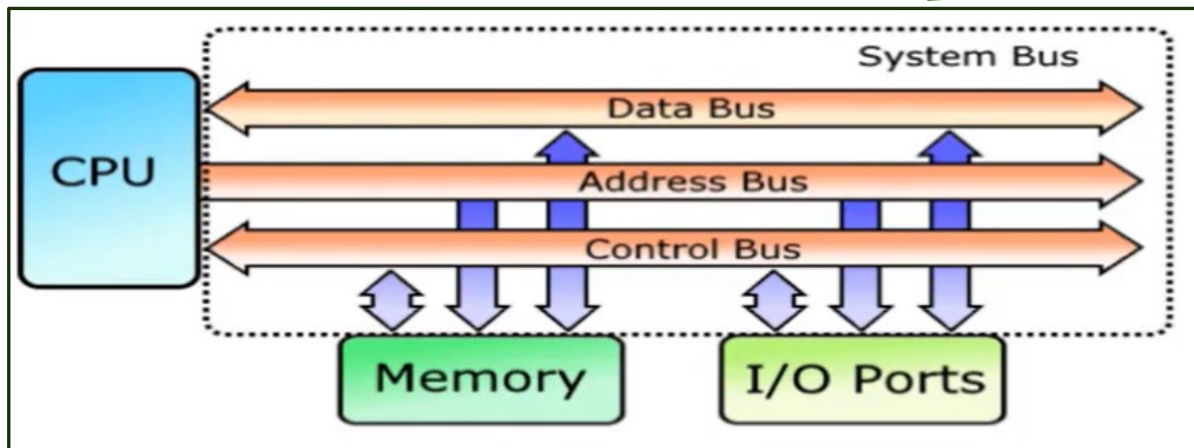


Types of Microcontroller

Microcontrollers: Types

➤ On the Basis of Bus Width

A single package



Bus Layout in Microcontroller

Microcontrollers: Types

➤ On the Basis of Bus-Width

- The bus width refers to the number of parallel lines or wires or connections.
- Increasing the bus-width of a microcontroller increases overall performance.
- On the basis of bus-width, the microcontroller are divided into:
 1. 8-bit
 2. 16-bit
 3. 32-bit

Microcontrollers: Types

➤ On the Basis of Bus-Width: 8-bit

- It can transfer and process the data of 8 bits in a single cycle.
- 8-bit ALU:
Needs multiple cycles for larger data size.
- 8-bit timer/counter

Microcontrollers: Types

➤ On the Basis of Bus-Width: 16-bit

- It can transfer and process the data of 16 bits in a single cycle.
- 16-bit ALU:
More efficient in performance comparing to 8-bit microcontroller
- 16-bit timer/counter
0x0000 (0) to 0xFFFF (65536)

Microcontrollers: Types

➤ On the Basis of Bus-Width: 32-bit

- It can transfer and process the data of 32 bits in a single cycle.
- Expensive and consume a lot of power.
- Complex tasks: signal and image processing

Microcontrollers: Types

➤ **On the Basis of Memory: Embedded Memory**

- Fixed size of memory blocks-Compact size
- Some microcontrollers have the feature of supporting external ROMs

Microcontrollers: Types

➤ On the Basis of Memory: External Memory

- Does not have one of the essential memory blocks inside its chip
- The use of external modules increases the size of the overall device

Microcontrollers: Types

➤ **On the Basis of Instruction Set Architecture (ISA)**

ISA includes addressing modes, instructions, data types, registers, interrupts, and external I/O:

Read/write, copy/move, add/subtract, multiply...

- **CISC (Complex Instruction Set Computer)**
- **RISC (Reduced Instruction Set Computer)**

Microcontrollers: Types

➤ On the Basis of ISA: CISC

- Large size of instruction set
- Many addressing modes
- Longer time to perform a command
- Parallel execution of an instruction is not possible in CISC

Microcontrollers: Types

➤ On the Basis of ISA: RISC

- Simple instructions
- Reduced size of instruction set
- Many line of codes are needed for a program
- ...

Microcontrollers: Types

➤ On the Basis of Architecture

A set of rules that defines the functionality and implementation of a computer (not a CPU):

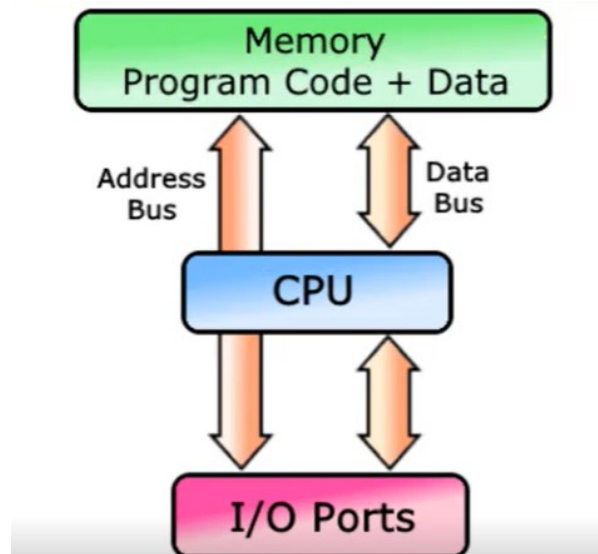
- Harvard Architecture
- Von Neuman Architecture

Microcontrollers: Types

➤ On the Basis of Architecture: Von Neuman

- The most used architecture in all computers
- Uses a single memory for program and data
- One bus is used for data and fetching instructions:

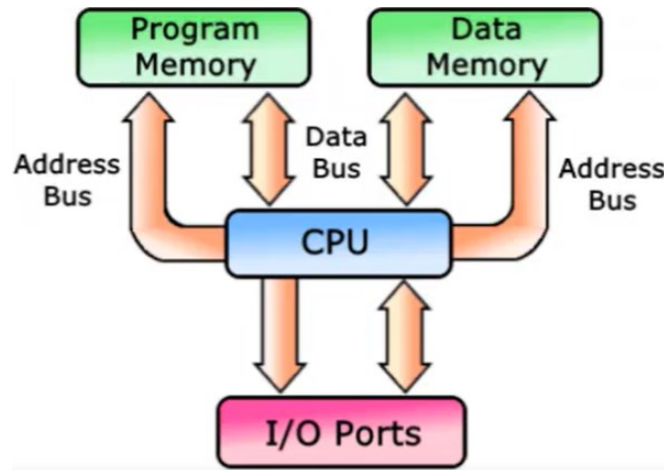
Both operations cannot be done at the same time (takes two machine cycles to complete an instruction).



Microcontrollers: Types

➤ On the Basis of Architecture: Harvard

- Has physically separated memory for instructions and data (Program memory and Data memory)
- Bus lines can be accessed both at the same time
- Can complete an instruction in one machine cycle



Microcontrollers: Types

➤ Generally used Microcontrollers:

- PIC Microcontroller
- 8051 Microcontroller
- AVR Microcontroller
- ARM Microcontroller
- ...

Microcontrollers: Types

➤ Microcontroller's Manufactures:

- Microchip
- NXP
- STMicroelectronics
- Texas Instruments
- ...

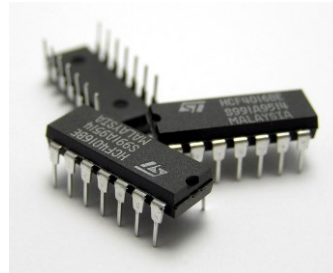


Microcontrollers: Types

➤ Packaging

■ Dual-in-line Package (DIP)

It is the most common through-hole IC package used in circuits, especially hobby projects. This IC has two parallel rows of pins extending perpendicularly out of a rectangular plastic housing.



■ Surface-mount Device (SMD)

SMD IC packages usually need custom PCBs, containing a matching pattern of copper on which they are to be soldered. Usually, special automated tools are used to solder these on PCBs.



Microcontrollers: Types



STM32 portfolio

 MPU					STM32MP1 Up to 1 GHz Cortex-A7 209 MHz Cortex-M4		STM32MP2 Dual 1.5 GHz Cortex-A35 400 MHz Cortex-M33		
 High-performance MCUs					STM32F2 Up to 398 CoreMark 120 MHz Cortex-M3	STM32F4 Up to 608 CoreMark 180 MHz Cortex-M4	STM32F7 1,082 CoreMark 216 MHz Cortex-M7	STM32H7 Up to 3,224 CoreMark Up to 600 MHz Cortex -M7 240 MHz Cortex -M4	STM32N6 3,360 CoreMark 800 MHz Cortex -M55 Neural processing unit
					STM32F3 245 CoreMark 72 MHz Cortex-M4	STM32G4 569 CoreMark 170 MHz Cortex-M4	<i>Mixed-signal MCUs</i>		
 Mainstream MCUs	STM32C0 114 CoreMark 48 MHz Cortex M0+	STM32F0 106 CoreMark 48 MHz Cortex-M0	STM32G0 142 CoreMark 64 MHz Cortex-M0+	STM32F1 177 CoreMark 72 MHz Cortex-M3					
	STM32L0 75 CoreMark 32 MHz Cortex-M0+	STM32U0 140 CoreMark 56 MHz Cortex-M0+	STM32L4 273 CoreMark 80 MHz Cortex-M4	STM32L4+ 409 CoreMark 120 MHz Cortex-M4	STM32L5 443 CoreMark 110 MHz Cortex-M33	STM32U5 651 CoreMark 160 MHz Cortex-M33			
 Ultra-low-power MCUs					STM32WL 162 CoreMark 48 MHz Cortex-M4 48 MHz Cortex-M0+	STM32WB0 64 MHz Cortex-M0+	STM32WB 216 CoreMark 64 MHz Cortex-M4 32 MHz Cortex-M0+	STM32WBA 407 CoreMark 100 MHz Cortex-M33	
 Wireless MCUs									
									



Latest product generation
 Radio coprocessor only
 New series or lines introduced in 2024

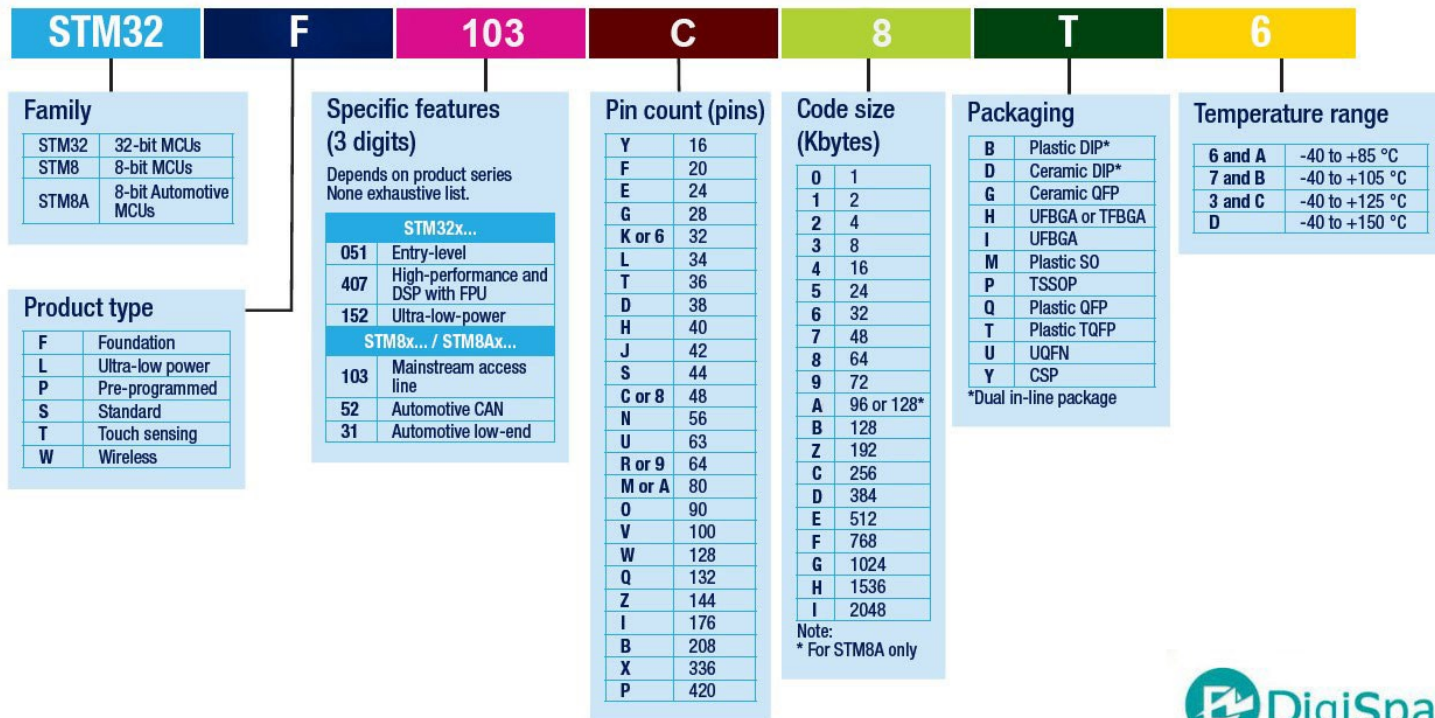


Source: OMDIA CLT, 2022, 2023

Microcontrollers: Types

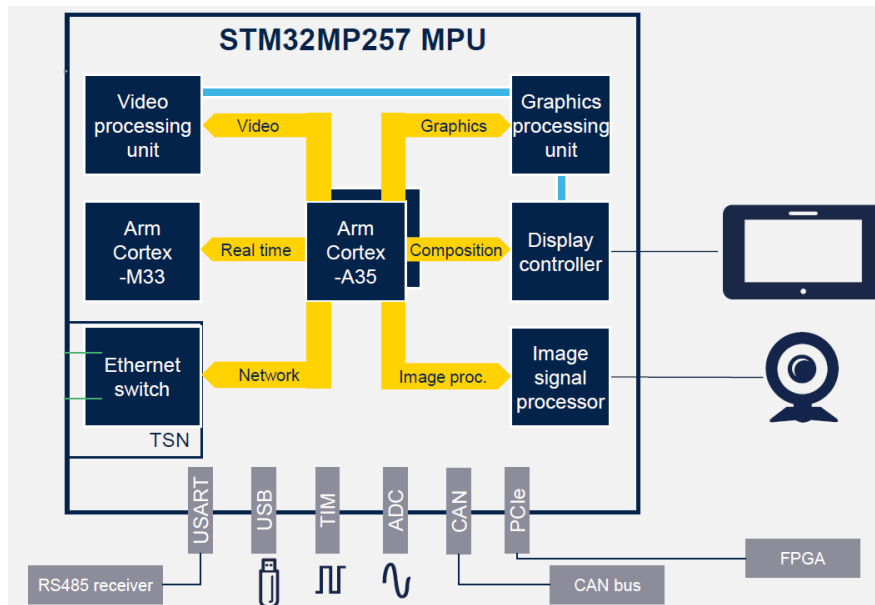
➤ STM32

MCU – Typical designations and part number suffixes



Microcontrollers: Types

- STM32MP257 application processors (STM32 MPUs) are based on a dual Arm® Cortex®-A35 cores running at up to 1.5 GHz and Cortex®-M33 cores running at 400 MHz.
- They feature 3 Ethernet ports, 3 FDCAN interfaces, H.264 encoding/decoding, a 3D GPU, AI/NN capabilities, LVDS/DSI display interfaces, and security features.
- STM32MP257 offers a dedicated evaluation board, the STM32MP257F-EV1, to help you get started with your design.

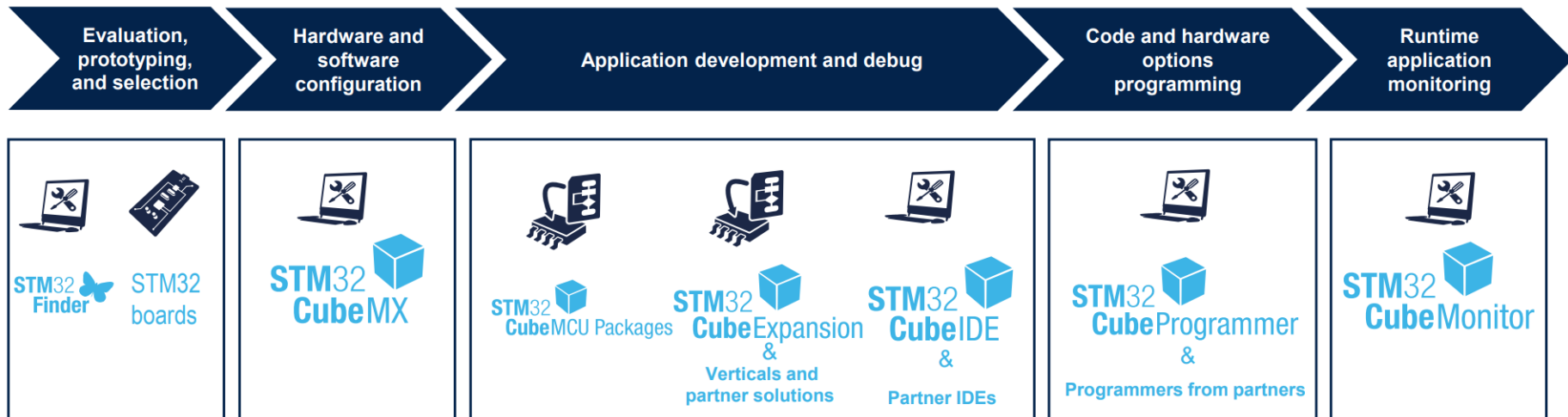


The STM32MP257F-EV1 Evaluation board



Microcontrollers: Types

➤ STM32Cube framework



Microcontrollers: Types

➤ Drivers

LL drivers

Lower abstraction level

Lower code size

HAL drivers

Higher abstraction level

Higher portability and reuse